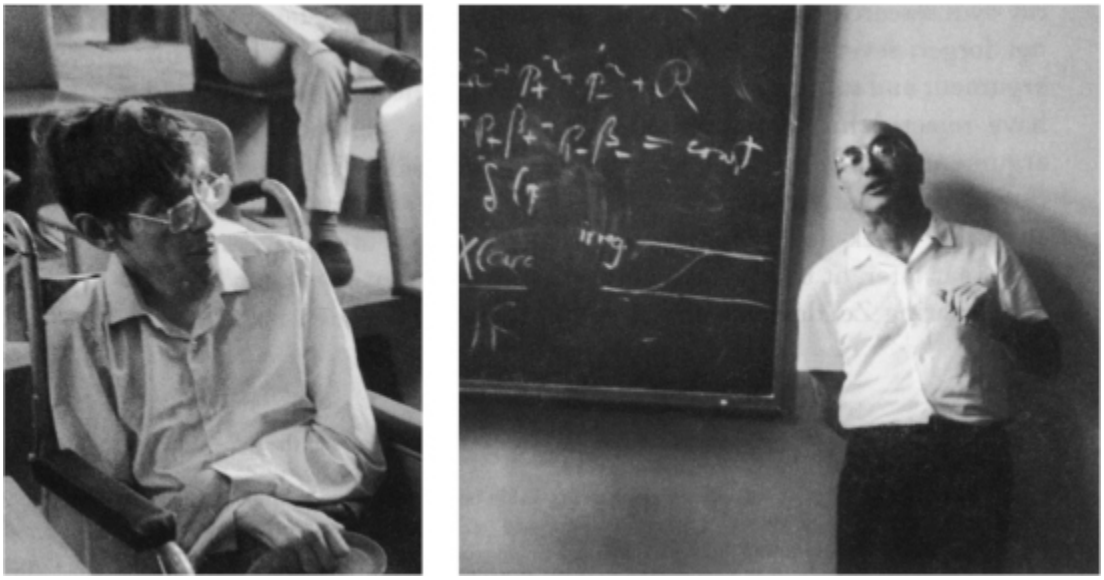


Left: **Stephen Hawking listening to a lecture at the Les Houches summer school in summer 1972.** ***Right:*** **Yakov Borisovich Zel'dovich at the blackboard in his apartment in Moscow in summer 1971.** [Photos by Kip Thorne.]

Of all the things Hawking learned from his conversations in Moscow, this one intrigued him most. However, he was skeptical of the manner in which Zel'dovich and Starobinsky had combined the laws of general relativity with the laws of quantum mechanics, so, after returning to Cambridge, he began to develop his own partial marriage of quantum mechanics and general relativity and use it to test Zel'-dovich's claim that spinning holes should radiate.

In the meantime, several other physicists in America were doing the same thing, among them William Unruh (a recent student of Wheeler's) and Don Page (a student of mine). By early 1974 Unruh and Page, each in his own way, had tentatively confirmed Zel'dovich's prediction: A spinning hole should emit radiation until all of its spin energy has been used up and its emission stops. I would have to concede my bet.

Black Holes Shrink and Explode

Then came a bombshell. Stephen Hawking, first at a conference in England and then in a brief technical article in the journal *Nature*, announced an outrageous prediction, a prediction that conflicted with Zel'dovich, Starobinsky, Page, and Unruh. Hawking's calculations